Legacies in the Lines:
The Lasting Walkability Impacts of
Portland, Oregon's Redlined Neighborhoods

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Abstract

This study examined whether historical redlining by the Home Owners' Loan Corporation (HOLC) in the 1930s continues to shape walkability in Portland, Oregon. Using regression models, walkability scores from Portland's Complete Neighborhoods index were compared across 68 neighborhoods assigned HOLC grades A - D, with vulnerability and race as controls. Contrary to the hypothesis that formerly redlined neighborhoods would exhibit lower walkability, results showed a statistically significant negative relationship: Redlined neighborhoods scored about 20–46 points higher than Grade A neighborhoods across all models. These findings suggest that historic racist policies impact modern walkability outcomes, but that Portland's reinvestment efforts could be impactful. The study also found that higher concentrations of people of color predict lower walkability within grade-equivalent neighborhoods, confirming that walkability remains an inequitably distributed resource.

Introduction

The United States has a long and well-documented history of using policy to determine how people are spread within cities, and who is allowed access to desirable neighborhoods. A prominent example of this segregationist policy is redlining. The federal government made maps of US cities where neighborhoods were graded from A-D, with D, or redlined areas, being low-income neighborhoods with high minority populations. These ratings were used by the government to make financial decisions and loan approvals, with redlined areas marked as financially risky or unworthy of federal help.

My study examined how redlining from the 1930's shapes walkability today in Portland, Oregon. First, I established that walkability improves social connections, physical health, longevity, and mental well-being. The literature showed that walkability can be an incredible tool to improve a community's quality of life. Previous research also showed how black people and other minority groups tend to have worse health outcomes despite being more likely to live in dense urban areas, which are thought to be more walkable. Next, I examined redlining and the impact it had on the financial development of cities. Redlined areas have long history of being denied monetary support from both commercial banks and the federal government. Furthermore, previous research connected redlining with a general lower quality of life. My paper considered whether redlining could explain why black people have worse health outcomes than expected, given that they typically live in areas with higher walkability rates. I hypothesized that neighborhoods in Portland that were previously redlined would have lower rates of walkability than those that were not redlined.

To test the hypothesis, I assigned each neighborhood in Portland a walkability score and redlining grade based on the original mapping. I conducted a series of linear regressions treating redlining as both a binary variable and an ordinal variable to assess if there is a relationship specific to redlined areas or a general connection between previous neighborhood grades and walkability.

The impacts of redlining may be strongly shaped by the historical context of individual cities. My paper focused on Portland because it has a reputation for leading the way in walkability policy, while also having a history of strict racial segregation that has uniquely shaped how black people experience living in Portland. The following section provides an overview of Portland's policy and development that impacted black people and other minorities. This history will clarify why Portland offers a compelling case for examining the relationship between redlining and walkability.

Context Chapter - Portland

For most places in the United States, political racism has shaped how and where people live and experience their lives within the country. Since before it was officially a state, Oregon, and the city of Portland, have placed restrictions on where minorities are legally able to live. This study uses people of color and minorities to represent those impacted by racist policies, although it should be noted that many racial and ethnic groups were targeted by these policies, including Jewish and Italian people (Hughes et al, 2019). Black people were the majority group and most significantly impacted by redlining in Portland, so this study uses black people as the group of focus. The 1844 'Black Exclusionary Law' made it illegal for black people to settle in the territory of Oregon. This law was not removed from the state's constitution until 1926 (State of Oregon). Due to the state's hostile environment for non-white people, including 9,000 Ku Klux

Klan members, the black population in Portland stayed small for decades, around 2,000 people or less than 1% of the total population, until World War II (Oregon Population, 2017). Post-Pearl Harbor, Portland became a center for ship building but lacked the labor supply to support this new industry (Geiling 2015). Within a few years, the black population grew to 20,000 people (Semuels, 2016). This sudden influx of people in Portland created a pressing need for housing, but the Portland government was not willing to create permanent housing for fear that black workers would remain after the war effort, and so was created the largest public housing project in America at the time: Vanport, Oregon. Housing between 20,000 and 40,000 people, over 6,000 of whom were black, Vanport had one of the highest population of people of color per capita in the Northwest. It was also the second most populated city in Oregon. Vanport was built in just 110 days on marshlands across the Columbia River from Portland (Geiling, 2015). On Memorial Day in 1948, water broke through the dikes protecting the city, and decimated Vanport. The housing authority of Portland (HAP) gave no warning of the floods to Vanport residents, despite clearing Vanport HAP offices of official documents, suggesting previous knowledge of the floods. Within less than a day, the entire city was destroyed leaving over 6,000 black residents with nowhere to live.

After the destruction of Vanport, there was only one place in Portland where black people were able to move: the Albina neighborhood. Prior to WWII, one of the only jobs black people were legally allowed to work was as a railroad porter, so Portland's black residents settled in Albina, within walking distance of Portland's Union Station (Geiling, 2015). Later, Albina was the only option for displaced black people due to segregation, racism, and affordability.

Vanport would not be the last time the black community in Portland was displaced. In the early 1960's, Portland started an urban renewal project (History of Albina, 2023). These plans

involved the construction of the I-5, the expansion of a hospital, and the building of the city's biggest stadium. To make room for these city expansions, they bulldozed through the middle of the Albina neighborhood, displacing hundreds of black families and businesses. Once again, many black people were forced to start from nothing, continuing the prevention of black generational wealth. At the time, the city of Portland celebrated this action as a positive expansion of the city, with little regard for the lives it would destroy. Black people in Portland have carried that betrayal with them to the present day. As the city looks to make progressive planning decisions, it is often met with justified mistrust and opposition from black Portlanders looking to preserve their homes.

In 2010, Portland Bureau of Transportation (PBOT) announced a simple plan to construct a bike lane through the heart of Albina (Miller and Lubitow, 2015). This plan would spark heated debate between white bike riders and black North Portlanders. Biking had come to be a symbol of gentrified neighborhoods and people were worried that these lanes would lead to increased gentrification. Additionally, black Portlanders were frustrated at being left out of the conversation, especially when they had been asking for years for increased pedestrian safety protections that had been ignored. The city was surprised to be met with such strong opposition because it had not considered its own historical actions that would lead to the alienation of its black citizens.

In more recent years, Portland has been taking actions to combat its racist systems that have existed since the city's conception. In 2014, the Portland Housing Bureau introduced a $70 million housing strategy for Albina residents impacted by economic discrimination. They have also increased community-based committees that involve members of historically black neighborhoods to be decision making parters in city planning (Ali 2025). These recent policy efforts highlight Portland's recognition of the long-term harm caused by its racialized planning

decisions. This context highlighted a need to examine modern policies through their impact on black people and provided the context for which this study will examine walkability and redlining.

Literature Review

Walkability has become one of the most popular qualities in urban environments, reflecting the growing research on its benefits for health, accessibility, and overall quality of life (Five Reasons, 2015). On the other hand, research on redlining exposes how discriminatory lending practices produced systemic patterns of disinvestment and inequality. When considered together the central question is raised: how do historical patterns of exclusion influence the conditions that lead to walkability today? The following literature review explores key findings of walkability and redlining to establish the theoretical expectations that guide my study.

Walkability

Walkability is a relatively recent concept emerging from a wave of urban critique in the 1960's. Jane Jacobs, with her paradigm-shifting work, 'Death and Life of Great American Cities' brought up new views of modern urbanism, specifically car-centric planning and how it inhibited the pedestrian's experience (Ali 2025). While it would be decades before these ideas would lead to a formal definition, it was Jacobs' framing of cities as a pedestrian experience, that would lead to our modern view of walkability.

Chris Bradshaw is commonly cited as the first to formally define walkability. In his 1993 paper, he noted a need for an index of neighborhoods that could be used to collect property taxes, create initial development fees, and for homebuyers to make buying decisions. Although future works often dissect the social and physical health aspects of walkability, it is important to note

that since its creation, walkability was designed as a monetary tool for home purchasing. Bradshaw proposed four basic characteristics to define walkability:

1) A "foot-friendly" physical environment,
2) Access to 'useful' destinations by foot,
3) A natural environment that moderates the weather,
4) A social and diverse culture.

Bradshaw goes on to create the first index for walkability. Bradshaw's definition highlights social and diverse cultures (diverse was here left undefined; its meaning thought to reflect a variety of neighborhood activities not demographic diversity) and his scale includes subjective measures such as "women's rating of neighborhood safety" and number of "places of significance" to the respondent. These subjective measures are essential to this original definition of walkability as they use the resident's perception of their own home to assign value rather than being a blind tool that performs a mathematical function outside of the lived experience. Future models used for defining and measuring walkability often lose that aspect of perspective.

Since the early 2000's, many more walkability and pedestrian guidelines have been created, and they capture a wide variety of measurements. Some guides reflect Bradshaw's view of walkability as a person's perception of their environment. For example, the National Highway Traffic Safety Administration assesses walkability through a perceived measure of safety (NHTSA, 2024). Guidelines that are used in broad contexts are limited by the cost and time to employ measurements. The Environmental Protection Agency (EPA) created the National Walkability Index as a standard to compare walkability among US cities (Thomas and Zeller, 2021). The guide includes a mathematical index that measures aspects such as intersection density and proximity to transit stops. The EPA's guide uses specific facts that can be calculated

through online mapping. Similarly, in 2007, Redfin, a real estate service provider, launched Walk Score. Measuring distances to amenities, transit, paths, and more, Walk Score gives a value to addresses and cities that is representative of how easy it is to walk or live without a car (About Walk Score, 2019). Walk Score is commonly used today by researchers to define walkability because it can be easily calculated for any city throughout the US.

While these objective measures make walkability easy to measure in a uniform way over large areas, they lose nuances that come from close assessment. For example, two streets might have the same number of sidewalks, and therefore produce the same score according to a guideline, but one street could have wheelchair accessible ramps, tree shade, garbage cans, and benches while the other street has a crumbling curb and few marked crosswalks. People in the first scenario might choose to walk more because they feel safer and happier. These differences could make significant changes in access, equity, and quality of walkability, but that would not be captured in a simple objective measurement of sidewalk density.

The City of Portland, Oregon was one of the first to create an official guideline for walkable cities in 1994 and later published in their 1998 ‘Pedestrian Master Plan’ (City of Portland). This plan was groundbreaking in many ways. It highlighted walking as transportation, allowing state and city transportation agencies to provide support for the creation of a walkable city. The plan lays out seven design principles for pedestrian improvement including safety, ease, and connection as key parts of a walkable city. The document also analyzed the existing design of Portland finding certain areas ‘deficient’. Following this rating, the plan proposed project areas for parts of the city that need pedestrian improvement. According to their graphs, the suburban outskirts of the city were in most need of improvement, while the inner-city center has the most walkability, so they originally focused their investment on suburban outskirts. Portland,

Oregon continued establishing itself as the front lines of walkability study with constant updates and creations of pedestrian scores (Duarte et al., 2025). In popular culture, Portland is recognized for its walking, and specifically, bike friendliness. Condé Nast named it in the top 5 most walkable cities in the US, Walk Friendly Communities gave Portland the highest designation for being walkable, and PeopleForBikes ranked it in the top 10 most bikeable cities (Towey, 2024; Strategies, 2017; 2022).

It is only within the past 20 years that research on walkability has been expanded to include a variety of cultural, social, and physical health outcomes. A 2005 study found that measures of mixed land use, walkable infrastructure, and residential density were positively associated with the amount of physical activity a person has in their day (Frank et al., 2005). When it is easier and safer for people to walk, they are more likely to choose it over cars for transportation. This increase in activity in walkable communities reduces chronic diseases and specifically supports older adults in being more independent (Kerr et al., 2012). Current research suggests that walkability plays an important role in the complex social relationships and communities of the American city. Walkability is positively related to social interactions with neighbors and a sense of community (Carson et al., 2023). People who live in walkable neighborhoods have more social capital. They are more likely to participate socially and politically than people who live in suburbs or car-dependent areas (Leyden, 2003). Additionally, walkable neighborhoods are inversely related to depression symptoms (Berke et al., 2007). The literature suggests that walkability could significantly improve almost every aspect of quality of life. For one element of city design to benefit people is such wide-reaching ways should not be taken lightly, as it has the potential to significantly improve people's lives.

Although there is evidence that walkability improves health outcomes, uncertainty remains as to how tightly the two are connected, and what other factors could be contributing to this relationship. A study on the health outcomes of urban design found that high population density, mixed land use, and greater transit access was associated with a lower BMI among rich, educated white people. This same association was much less consistent among disadvantaged groups (Lovasi et al., 2009). It is also well documented that black people have a significantly higher risk for cardiovascular disease than white people, which contradicts previous knowledge walkability benefits given that white people are more likely to live in less walkable areas (Zakeri et al., 2021.). Together these findings point out a glaring hole in our understanding of walkability, as people of color, and black people specifically, face adverse health issues that walkability supposedly addresses.

A 2025 study on mental health and walkability found that mental health encounters with the police were three times higher in areas with high walkability compared to less walkable ones. Because this opposes previous claims that walkability improves mental health, researchers found a need for "a deeper understanding of how demographics, socioeconomic factors, and neighborhood characteristics interact (Makram et al., 2025)." The previous 2007 study on walkability and depressive symptoms noted that their research did not include racial or ethnic composition and therefore needs further work into racially diverse settings (Berke et al., 2007) These studies acknowledge that racial demographics might shape walkability outcomes, but this factor has not been thoroughly explored. While it is known that racial segregation reduces the positive association between health and walkability, many aspects of discrimination that likely impact walkability outcomes are unstudied (Huang, 2022). Understanding why black people are less likely than white people to experience the positive outcomes of walkability, will require a

deeper understanding of how historical systemic racism impacts modern ideas, policies, and relationships. The next section of the literature review shows how redlining is one such racist policy that caused systemic discrimination against people of color specifically by preventing the improvements of homes and neighborhoods.

Redlining

Redlining is a historical practice that emerged from the New Deal in the 1930's as a response to the Great Depression. Under President Rosevelt, the Home Owner's Loan Corporation (HOLC) was created to support home owners at risk of losing their homes to foreclosure. The HOLC developed a housing plan that served as the basis of the National Housing Act of 1934. This act created the Federal Housing Administration (FHA) with the purpose of providing loans and creating a mortgage fund to help Americans buy houses (National Housing, 1934). The goal of the FHA was to insure banks and mortgage companies to spur the building of new homes and encourage construction and repair of existing buildings ("FDR and Housing", 2012). From their origin, both the FHA and the HOLC established that areas with high populations of black people or other minorities were high risk, so banks and mortgage companies were not willing to invest in them. The FHA Manual, which instructed workers on how to decide who is eligible for loans, states that location ratings are threatened by the "probable infiltration of inharmonious racial groups (Underwriting Manual, 1938)". During the 1930's, banks were under heavy restrictions when providing mortgage loans as the Federal Reserve feared that locking up funds in mortgages could create another financial crisis. Due to the lack of private loans, about half of mortgages during the 1930's and early 40's were financed by the FHA, and therefore largely inaccessible to people of color (Rose, 2023). The HOLC created 'residential security' maps of cities in the United States which were then used by both the HOLC and the FHA in

deciding how to distribute loans. Neighborhoods were ranked from 'A' to 'D' meaning "Best", "Still Desirable", "Definitely Declining", and "Hazardous". Hazardous or D was colored red, leading to the term redlining. Hazardous areas were deemed high risk, and lending was discouraged to these areas. The FHA, as well as commercial banks and lenders, created their own maps to make financial decisions, but the FHA destroyed most of its maps so few survive today (Nelson, 2023). It is difficult to determine which decisions used which specific maps, but today the term redlining typically refers to the broad practices that the FHA and the HOLC used to separate space by racial distributions in order to discriminate against black people and minorities by depriving them of the same financial supports that were much more readily available to white people (Mitchell and Franco, 2018).

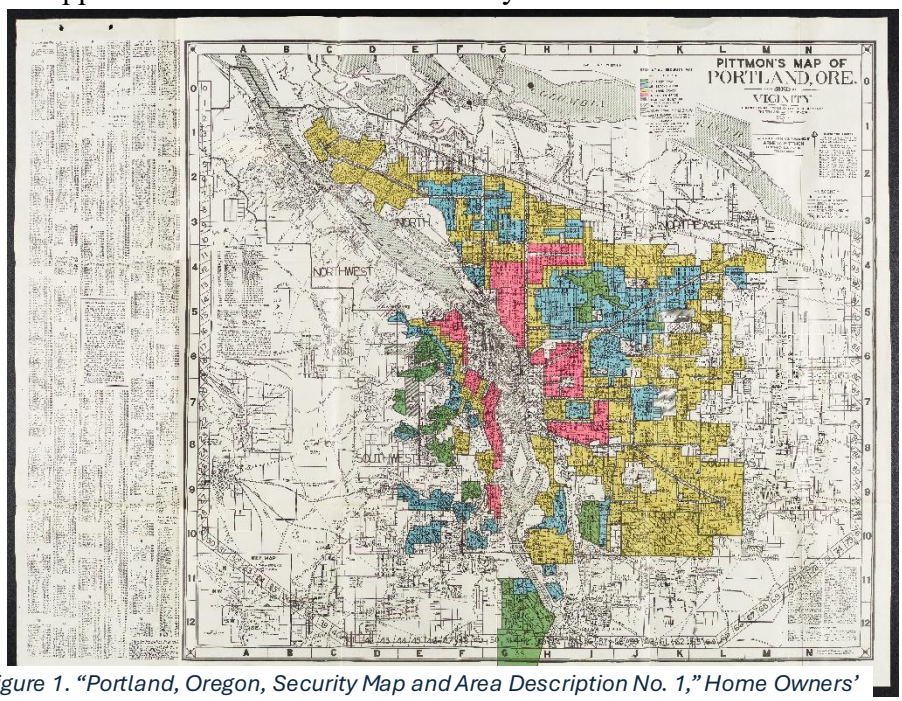


Figure 1. "Portland, Oregon, Security Map and Area Description No. 1," Home Owners' Loan Corporation, National Archives

The practice of redlining was not made explicitly illegal until the 1968 federal Fair Housing Act. The act prohibited discrimination of people based on race, color, national origin, and religion when selling or renting housing. It required cities that receive federal funding to "affirmatively further fair housing", meaning that cities must promote fair housing and take actions to overcome historic segregation and discrimination (Hughes et al., 2019).

Despite being outlawed in the 60's, Redlining has caused black and minority populated neighborhoods to be decades behind in wealth building and financial investment in their built

environment. Huang theorizes that "historical disinvestment in segregated neighborhoods could reduce the availability and accessibility of walkable neighborhood amenities (2022)." Disinvestment, or funds being withheld or barred from areas that would otherwise be eligible to receive them, could impact walkability in many ways. Tree canopy, the amount of area in the sky that is covered by tree tops and essential for climate control, environmental protection, and mental enjoyment of walkability, is associated with HOLC grading (Jung et al., 2024). Areas with higher grades have lower surface temperatures and larger canopy areas. Redlining is associated with lower life expectancy, higher rates of poverty, and increased childhood obesity (Ali, 2025; Kowalski et al., 2023). There is a clear link between a lower quality of life and redlining, although the precise mechanism that causes this relationship is less understood.

Along with the lasting effects of initial redlining in the 1940's, there is evidence that redlining continued well after the 1968 Act. In the 1990's, a local Portland newspaper published the 'Blueprint for a Slum' investigation into redlining which found that major mortgage lenders discouraged home ownership in North and Inner Northeast Portland, otherwise known as the Albina neighborhoods, and the same ones designated as red in the HOLC maps (The Oregonian). Specifically, the investigation concludes that lenders did not make any federally insured mortgages in the census tracts with the highest black population and they imposed minimum loan amounts that made many low-income neighborhoods ineligible for loans. For decades the black populated areas of Portland have been barred from financial aid that could help them develop their communities and neighborhoods. The Federal Community Reinvestment Act, a law passed with the intention of preventing urban decay, required commercial banks and financial institutions to reinvest in the communities they serve. However, by creating barriers for low-income areas to participate in financial services, these areas are not required to receive financial

reinvestment (Federal Reserve Board). The federal laws passed to end and redress redlining have not done enough to break the pattern of suppressing black neighborhoods through withholding financial support.

Our understanding of walkability and our understanding of where people live create a paradox. Walkability is associated with many positive health outcomes, yet socioeconomically disadvantaged populations and minorities are more likely to live in urban areas which are inherently more walkable (King and Clarke, 2014). The most common demographic in unwalkable, car-centric suburbs, is white people. They make up about 70% of all suburbs (Massey and Tannen, 2017). “There is substantial evidence to support a conclusion that people walk and cycle less for transportation in low-density, single-use suburban neighborhoods” and given that physical activity decreases the risk of a variety of adverse health conditions and are responsible for at least 200,000 deaths per year in the USA, people in the suburbs should have worse physical health outcomes (Frank et al., 2005). This leads to the question: Why do people of color have significantly worse health outcomes when they appear to have better access to walkable cities than white people? Due to the variability across cities in his study, Bereitschaft suggests that we must understand each city’s unique socio-historical context to begin understanding its relationship with walkability. My research is intended to explore one aspect of Portland’s historical policy to get a better understanding of how walkability is situated for citizens of Portland. The next section of my paper will set up my study design and tools of analysis.

Study Design

Theory and Expectations

Redlining created a cycle of generational poverty that specifically targeted black neighborhoods. Neighborhoods given grade 'D' by the HOLC are decades behind in generating wealth and neighborhood improvement. According to the city of Portland there has been "prolonged disinvestment" in the Albina area (Hughes et al., 2019). The city is responsible for creating bike lanes, bus stops, protected cross walks, and other infrastructure that makes walking and non-car transportation safe and accessible. A lack of city intervention should lead to a worsened built environment which is a crucial element of walkability. Ali shows that redlining predicts health disparities in Portland (2025). Given that walkability is associated with improved health and black people tend to have worse health outcomes, I expect that areas that were historically redlined will be less walkable than neighborhoods that were given high grades by the HOLC.

Hypothesis

My hypothesis is:

H: In comparison of neighborhoods, those that were redlined will be less walkable than those that were not redlined.

H_0: In comparison of neighborhoods, there is no relationship between redlining and walkability.

Research Design

To test my hypothesis, I will use a quantitative, observational, cross-sectional research design to examine the relationship between historical redlining and current neighborhood

walkability in Portland, Oregon. The unit of analysis is neighborhoods in Portland. I manually assigned each variable to neighborhoods using a map of neighborhood boundaries from the City of Portland's open data (2020a). Walkability is used as the primary outcome of interest because it directly ties physical infrastructure to health outcomes as a result of city development.

This is a cross-sectional study because my variable data was not available over multiple years. While the HOLC grades were assigned in the 1940's, they are being treated as a fixed historical condition that impacts the present. Due to this research design, I will not be looking to establish causality. It is possible, and likely, that many unmeasured factors explain both HOLC grade assignments and current walkability scores, but by examining the relationship between the two, it allows us to explore systemic patterns that still shape people's lives.

The geographic scope of this study is limited to Portland, Oregon for several reasons. Portland represents a mid-sized American city with a well-documented history of redlining. The City of Portland provides high-quality, block level data about walkability and vulnerability. Portland has a significant interest in walkability planning, innovation, and policy. Lastly, walkability and how it intersects with racial justice is a prominent debate within Portland's discussions of city planning (Miller and Lubitow, 2015).

My independent variable, redlining, comes from the original Home Owner's Loan Corporation map published in 1938 and stored at the National Archives. I used a version of the map digitized to a geo-spatial database by the University of Richmond (Nelson, 2023).

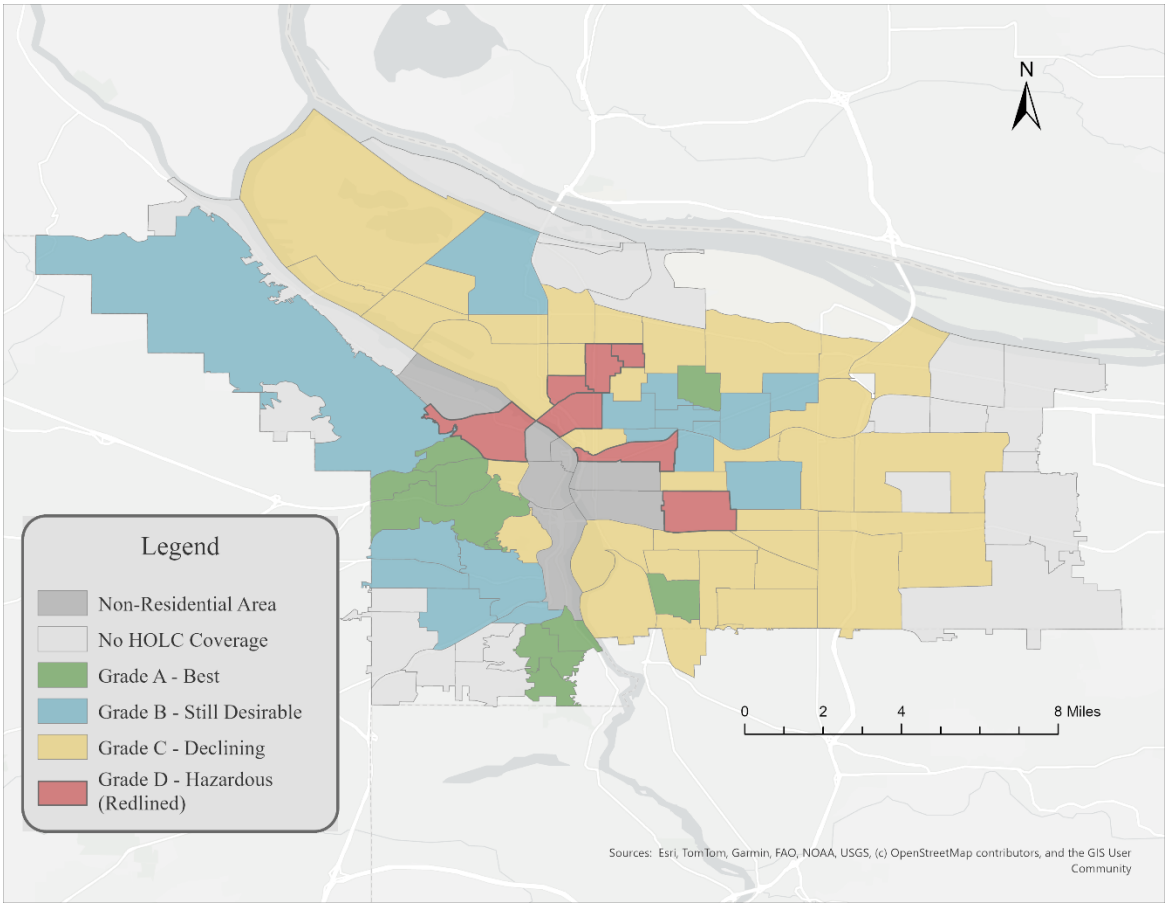


Figure 2 Portland Neighborhoods by HOLC grade

There were 94 HOLC designated polygons that were coded residential or non-residential. Only residential neighborhoods were given a grade from A-D. The original HOLC districts were not representative of modern neighborhoods, so I chose to use the HOLC grade that took up the largest percentage of each neighborhood. For example, if a neighborhood was 75% grade B, 15% grade A, and 5% grade D, it would be designated as grade B. Figure 2 shows my results after mapping HOLC grades to neighborhoods. There are 97 recognized neighborhoods in Portland. Twenty-two neighborhoods had no overlap with an original HOLC area, displayed in light gray, and were removed from the analysis, 7 neighborhoods, displayed in dark gray, were in previously industrialized areas, not given grades and therefore removed from the analysis. The remaining 68 neighborhoods were kept in the study. Grade C neighborhoods had the largest share

of the sample at 50% (n = 34), Grade B neighborhoods account for 25% of the sample (n = 17), Grade A for 14.7% (n= 10), and Grade D for the smallest share, 10.3% (n = 7). Spatially, Grade A neighborhoods tend to be in the West Hills and Southwest Portland reflecting the historically suburban character of these areas. Grade D neighborhoods are clustered in the inner Northeast side of Portland, which is expected given the historically large percentage of black and minority communities in these areas.

My dependent variable, walkability, comes from the City of Portland's complete neighborhoods map which was created in 2020 and last updated in 2024 (City of Portland, 2020b). Portland defined completeness as proximity to seven amenities:

1) Food,
2) Elementary Schools,
3) Transit,
4) Bike Routes,
5) Commercial Services,
6) Parks,
7) Pedestrian Infrastructure.

Each block area was given a score from 0 – 100, with 100 being most 'complete'.

The complete neighborhoods map was not designed to be specifically a measure of walkability, but I chose to use it for my walkability variable because it addresses the built environment, infrastructure, access to daily needs, transportation, and the natural environment. The combination of these elements allows us to examine if people can go about their daily lives without the use of cars, the core of walkability. Additionally, the data has 325,594 data points,

allowing for a very robust measure of walkability. When combining walkability with neighborhoods, I calculated each neighborhood's walkability score by multiplying the walkability value of each intersected piece by its area, adding all those products together, and then dividing by the total area. This gave me an area-weighted average, so that larger pieces of the neighborhood contribute more to the final score. The results are shown below in Figure 3. The scores ranged from 5.9 to 81.99, where 100 is a neighborhood that has complete easy walkable access to all amenities.

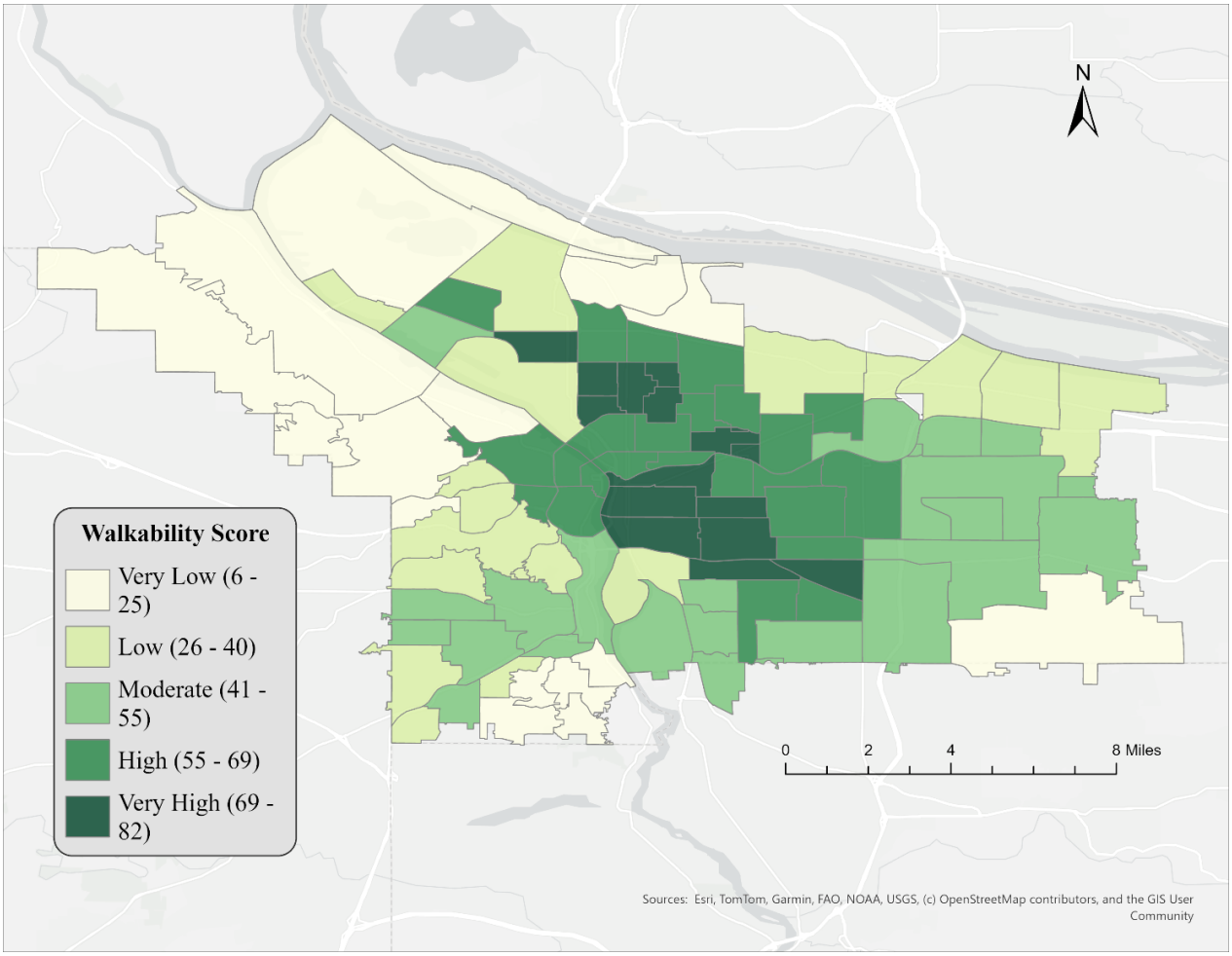


Figure 3 Portland Neighborhoods by Walkability

My control variable comes from the City of Portland's Vulnerability data set built by the Bureau of Planning and Sustainability (2020c). Portland identifies residents who are "more vulnerable to changing economic conditions" in this map. The variables used to determine vulnerability were the following:

1) Percent of people of color,
2) Percent housing cost-burdened (those who pay > 30% of income to housing costs),
3) Percent without college degrees,
4) Income level,
5) Households with limited English,
6) Percent of people with disabilities,
7) Percent of seniors (65+),
8) Percent of minors (21-).

The city combined and rescaled variables into a number from 0-100 for each area. Portland considers any area with a score of 60 or greater to be vulnerable. Once again, I assigned the scores to neighborhoods using an area weighted average that is mapped in Figure 4. I choose to use this variable as my control because it encompasses many ways in which people might face financial burdens or discrimination that could impact their home and neighborhood environment.

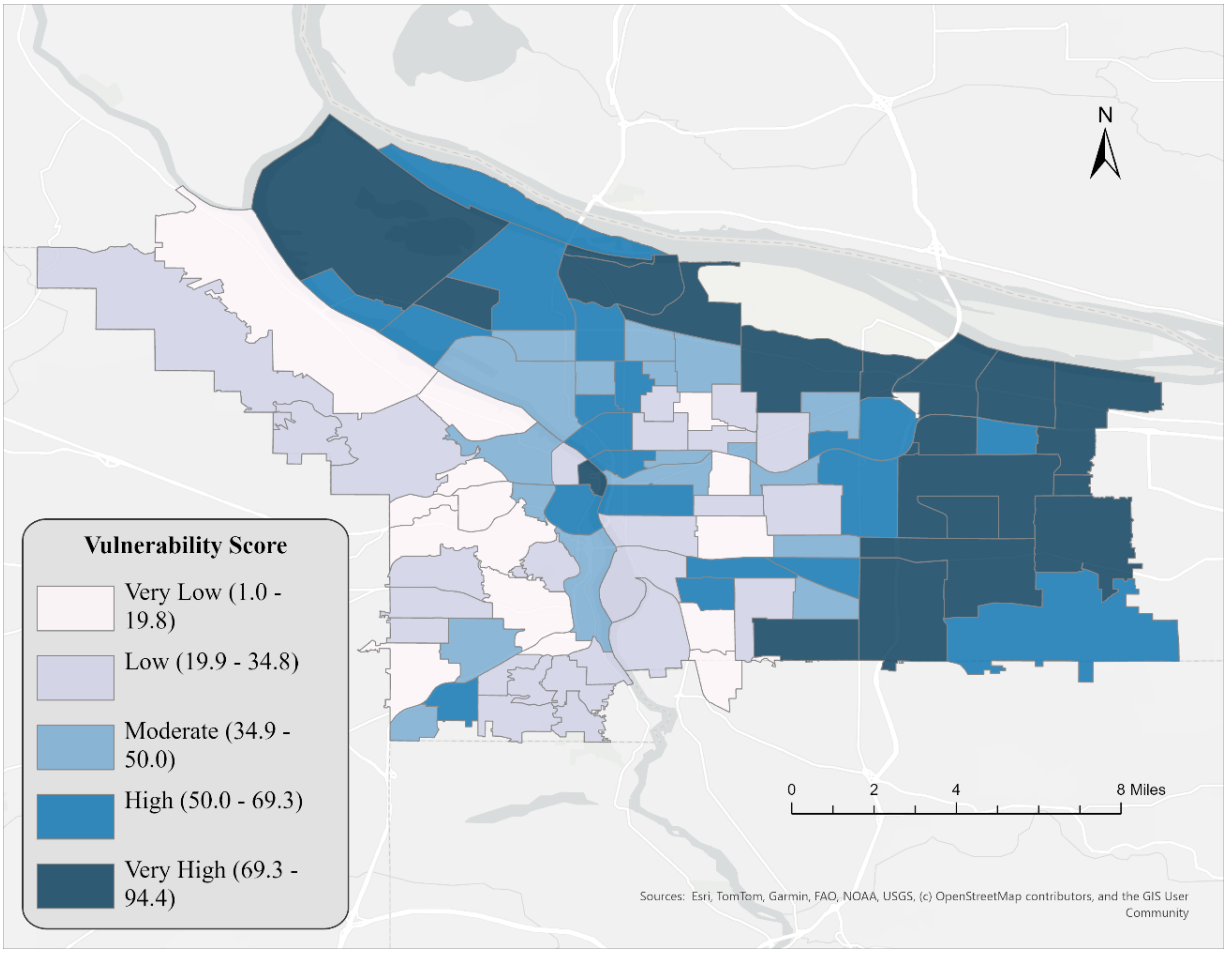


Figure 4 Portland Neighborhoods by Vulnerability Score

Operationalization

During my analysis I looked at redlining in two ways. I first treated it as a binary where grades of D were coded 1 to mean redlined and grades A-C were coded 0 to mean not redlined. However, there were only 7 neighborhoods graded D, so to create a stronger statistical finding, I ran an additional model where I treated the HOLC grades as categorical.

Variable Table:

Variable	Type	Source	Scale	Role
Walkability	Continuous	City of Portland	0-100	Dependent
Redlined	Binary	HOLC grades	0/1	Independent
Redlined Grade	Ordinal	HOLC grades	A/B/C/D	Independent
Vulnerability	Continuous	City of Portland	0-100	Control
Avg People of Color	Continuous	City of Portland	0-100%	Control
Avg Income	Continuous	City of Portland	$32,000-$125,000	Control

Table 1 Variables used in Data Analysis

Prior to finalizing my models, I examined percent people of color and average income by neighborhood to assess their relationship with the vulnerability variable. Area-weighted averages for both variables were calculated and mapped across Portland neighborhoods. These maps, listed in Appendix B, appear to be strongly consistent with the special patterns of vulnerability. Areas with high percentage of people of color and low income corresponded with areas graded highly vulnerable. These maps confirmed that all three variables would present high multicollinearity in a model design and would not be needed together in a model analysis. I selected vulnerability to use as my main control variables, although in model 3, I used percent people of color as an alternative control to assess the robustness of findings.

Regression Models:

Model	Equation	Notes
Model 1	$Walkability_i = \beta_0 + \beta_1(Redlined_i) + \beta_2(Vulnerability_i) + \varepsilon_i$	HC3 Robust Standard Errors
Model 2	$Walkability_i = \beta_0 + \beta_1(GradeB_i) + \beta_2(GradeC_i) + \beta_3(GradeD_i) + \beta_4(Vulnerability_i) + \varepsilon_i$	Grade A = Reference Category
Model 3	$Walkability_i = \beta_0 + \beta_1(GradeB_i) + \beta_2(GradeC_i) + \beta_3(GradeD_i) + \beta_4(PercentPOC_i) + \varepsilon_i$	Grade A = Reference Category
Sensitivity	$Walkability_i = \beta_0 + \beta_1(GradeB_i) + \beta_2(GradeC_i) + \beta_3(GradeD_i) + \beta_4(Vulnerability_i) + \varepsilon_i$	Forest Park Excluded (n = 67)

Table 2 Regression Models

Model 1 was the primary test of this study's central hypothesis. It operationalized redlining as a binary variable with vulnerability as a control. This model tested whether neighborhoods explicitly designated as hazardous by the HOLC differ from all other graded neighborhoods. Because heteroskedasticity was detected in this model, meaning the residuals do not have constant variance, HC3 robust standard errors are reported to ensure reliable interreference. If I were to find this model to be statistically significant and reject the null hypothesis, it would mean that redlined neighborhoods are directly associated with lower walkability. While a significant finding would not necessarily mean that redlining causes lower walkability, it could be understood that those neighborhoods continued to be underdeveloped and lacking proper planning and attention from the City of Portland. Future investigations into the health disparities between people of color and white people could assess how redlining is directly connected with modern health outcomes through the vehicle of walkability.

Model 2 provided information about variance across the full HOLC grade spectrum. Model 2 estimated separate coefficients for grades B-D all relative to grade A, which allowed me to assess if walkability changes incrementally as HOLC grades change or whether differences are marked at a particular grade threshold.

Model 3 expanded on Model 2 by substituting percent people of color for vulnerability scores as the control variables. By changing the control variable, I was able to test the robustness of the grade gradient with a second operationalization of neighborhood demographic conditions. It also allows this study to examine if racial composition predicts walkability after accounting for HOLC grades. Vulnerability was removed as a control due to its high collinearity with percent people of color. Model 3 has the best fit and is treated as the primary categorical model for interpretation. A statistical significance from either categorical model would show that the racist patterns created in the 1940's are still present today and have not been adequately addressed. These results would mean that each HOLC graded area is associated with a different level of walkability. Statistical significance would show that when defining, measuring, and creating walkability we must consider historical patterns of oppression and racist city planning as they will be correlated with walkability outcomes.

The last models, the sensitivity models, are Model 2 with one modification. They removed Forest Park, a B graded neighborhood, from the individuals. The neighborhood Forest Park is mostly made up of a city park by the same name. Forest Park is one of the largest urban forests in the US at over 5,200 square acres (Experience Forest Park). Due to the nature of the complete neighborhoods scoring index, the Forest Park neighborhood was given a very low walkability score (AVG_WALK = 5.9) as it does not have much infrastructure or many commercial amenities. Sensitivity Model uses vulnerability as a control while Sensitivity 1

Model uses average people of color as a control. The sensitivity models were run to ensure that findings from the previous model were not skewed by the inclusion of the forest park neighborhood.

Data Results

Analysis

The results from my analysis are contrary to my initial hypothesis. The relationship found is negative, meaning that as HOLC grades decrease (with A being best, or a higher grade, and D being worst, or a lower grade), walkability increases on the scale from 0-100, where 100 is highest walkability. This finding is consistent across both the binary and categorical operationalizations of redlining and is robust to alternate control variables. Table 3 presents the results of all my regression models. I will walk through the details of my analysis by each model below.

OLS Regression Models: Walkability and HOLC Redlining in Portland Neighborhoods

	Dependent variable:				
	Walkability Score (AVG				
	Model 1	Model 2	Model 3	Sensitivity	Sensitivity 1
	(1)	(2)	(3)	(4)	(5)
Redlined (D=1)	19.777***				
	(3.284)				
Grade B		27.156***	25.128***	30.586***	28.143***
		(6.532)	(6.296)	(6.051)	(5.899)
Grade C		31.675***	29.947***	32.281***	29.435***
		(7.039)	(6.095)	(6.443)	(5.647)
Grade D		45.856***	44.676***	46.300***	44.292***
		(8.489)	(7.958)	(7.768)	(7.371)
Vulnerability Score	0.019	-0.215		-0.232*	
	(0.102)	(0.113)		(0.103)	
Percent POC			-0.613*		-0.559*
			(0.232)		(0.215)
Constant	51.314***	35.716***	45.167***	36.043***	43.977***
	(5.347)	(5.548)	(7.155)	(5.077)	(6.636)
Reference Category	A/B/C=0	Grade A	Grade A	Grade A	Grade A
Forest Park Included	Yes	Yes	Yes	No	No
Observations	68	68	68	67	67
R^2	0.100	0.339	0.370	0.398	0.413
Adjusted R^2	0.072	0.297	0.330	0.359	0.375
Residual Std. Error	18.596 (df = 65)	16.188 (df = 63)	15.797 (df = 63)	14.811 (df = 62)	14.631 (df = 62)
F Statistic	3.594* (df = 2; 65)	8.064*** (df = 4; 63)	9.257*** (df = 4; 63)	10.260*** (df = 4; 62)	10.900*** (df = 4; 62)
Note:					*p<0.05; **p<0.01; ***p<0.001
	HC3 robust standard errors used for Model 1. Grade A is reference category for categorical models.				

Table 3 Model Regression Results

Model 1 concludes that the coefficient for Grade D is statistically significant, indicating that historically redlined neighborhoods have, on average, different walkability scores than non-redlined neighborhoods. Due to the low p-value ($p< 0.01$), I reject the null hypothesis. However, this model found a negative relationship between walkability and redlining, meaning that redlined tend to have higher walkability scores than non-redlined neighborhoods. On average, redlined neighborhoods score about 19.8 points higher on the walkability index than non-redlined neighborhoods after controlling for vulnerability. My analysis found that in comparison of neighborhoods, those that were redlined, will be more likely to be walkable than those not redlined. Vulnerability is not a statistically significant predictor in this model. Model 1 explains 7% of the variance in walkability score (Adjusted $R^2 = 0.072$), which is a low, but statistically significant fit.

Model 2 revealed a statistical association between walkability scores and redlining. Compared with Grade A neighborhoods, Grade B neighborhoods have a walkability score about 27 points higher ($\beta = 27.156$, $p < .001$), Grade C neighborhoods are about 32 points higher ($\beta = 31.675$, $p < .001$), and Grade D neighborhoods are about 46 points higher ($\beta = 45.856$, $p < .001$). Figure 5 below shows the spread of walkability by HOLC grade in which the linear relationship is clear. Grade A has the lowest walkability, and Grade D has the highest walkability with a very limited range. As seen in Figure 5, the relationship between historical grading is not random, as it follows a consistent gradient. Model 2 explains 33.9% of the variance in walkability scores ($R^2 = 0.339$), a substantial improvement from Model 1. Vulnerability was still not statistically significant in this model.

Even after adjusting for percent people of color, the linear pattern observed in the boxplot remains statistically significant in the regression results of model 3. The coefficients remain large

and positive. Grade B (β=25.128, p<0.001) and Grade C (β=29.947, p<0.001) have the smallest gap, while Grade D (β=44.676, p<0.001) remains the highest coefficient.

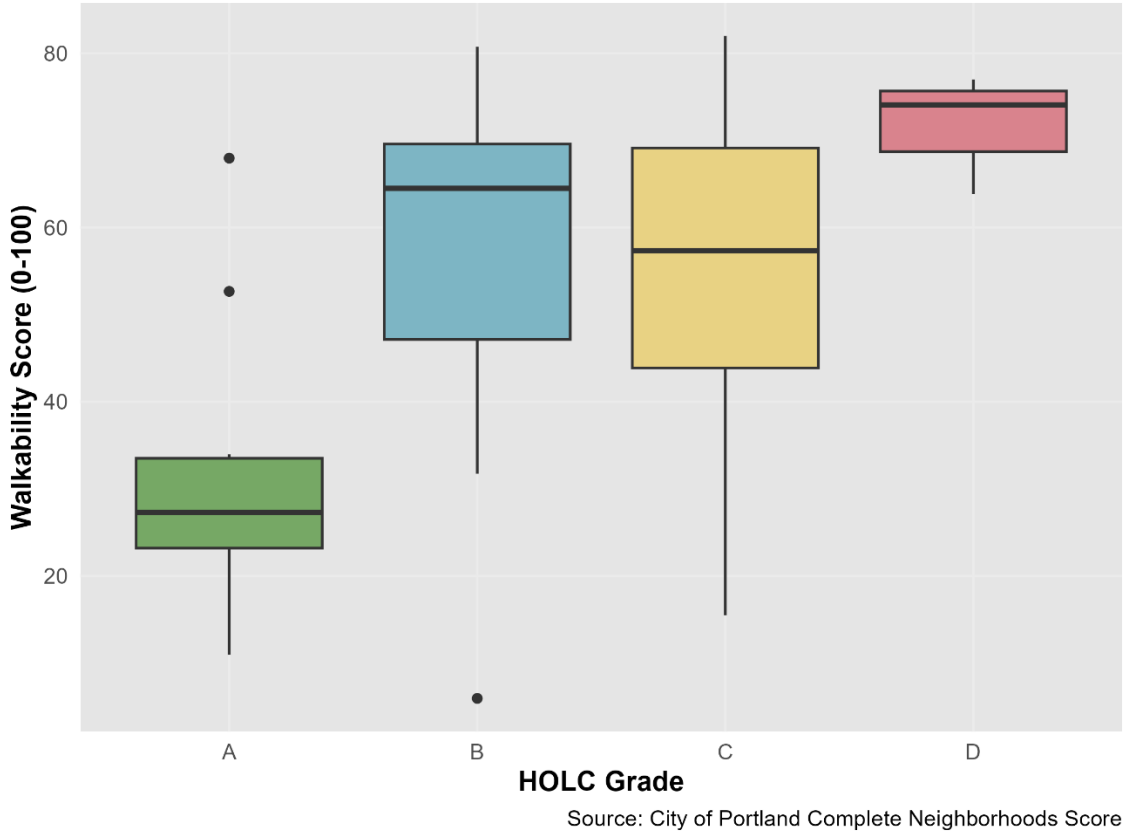


Figure 5 Spread of walkability by HOLC grade

Notably, from this analysis a negative coefficient (-0.613) between average people of color and walkability emerged as statistically significant with a p-value of 0.010. The negative coefficient indicates that neighborhoods with higher concentrations of people of color score lower on walkability after controlling for HOLC grade, revealing an important nuance in this study: while walkability and redlining have a negative relationship, within those grade districts, areas with a higher population of people of color are likely to have lower walkability. The statistical significance of average people of color warrants attention as vulnerability scores failed to reach statistical significance. Because the relationship was only significant for the people of color

variable, the data suggests that racial demographics capture a dimension of disadvantage that is not reflected in broader vulnerability indexes. Due to collinearity between these variables this study is unable to compare specific results between percent people of color and vulnerability, but the difference in significance suggests a further impact of racial inequality that should be studied further.

The sensitivity model replicates Model 2 with one exception, the removal of forest park (n =67). The relationship found earlier remains the same: Grade B (β =30.586, $p<0.001$), Grade C (β =32.281, $p<0.001$), and Grade D (β =46.300, $p<0.001$). The sensitivity results confirmed that the findings from the previous models are not driven by this outlier, but the vulnerability score reaches statistical significance in this model (β = -0.232, $p=0.028$). The coefficient is negative suggesting that after controlling for HOLC grade, more vulnerable areas have a lower walkability score. This model reveals that Forest Park could be responsible for the lack of significance with vulnerability in previous models. Sensitivity 1 Model uses average people of color as a control. The relationship between walkability and grades is not changed by a large amount. It is important to note that the average people of color coefficient (β = -0.559) and p-value (0.0118) are smaller than the vulnerability numbers which supports the previous finding that there could be a difference in how people of color experience walkability versus overall vulnerable people, even without the influence of Forest Park.

Altogether, the regression results consistently reject the null hypothesis of no relationship between redlining and walkability. However, they also reject this study's main hypothesis as the relationship between redlining and walkability is in the opposite direction of my original hypothesis. The implications of these findings are discussed in the following section.

Discussion

The results of this study do not support the primary hypothesis that formerly redlined neighborhoods would exhibit lower walkability. Instead across all models, neighborhoods designated as hazardous by HOLC appraisers and most directly associated with the denial of financial support, score significantly higher on the City of Portland's Complete Neighborhoods walkability index. The negative relationship between HOLC grades and walkability score, consistent among five model specifications and robust to alternate control variables, suggests that the decisions made when determining HOLC grades are involved in the characteristics that produce walkability scores today.

To understand this relationship, it is necessary to consider the historical context when the HOLC maps were created, and how suburbs were viewed. In the decades from the 1920's through the 1960's, car-dependent suburbs were idealized (Muller, 1977). Suburbs were a lavish escape from the dirt and crime of cities. Cars provided a new kind of freedom and access at the cost of metropolitan public transportation. The suburban sprawl still shapes the geography of cities today as the affluent outer edges remain car dependent. The HOLC graded down densely populated city centers because at the time, these areas were looked down upon, and only in the present day has the social lens switched to value dense, urban, and walkable areas (Bereitschaft, 2017).

While walkability is a positive element of communities, the takeaway from this paper should not be that low-income marginalized communities have better quality of neighborhoods than high-income white communities. Firstly, areas that were once redlined due to the large presence of minorities might no longer be accessible to those same communities. Gentrification is at the core of discussions about walkability and racism. As walkability has evolved to become

an essential component of home buying (the most common tool for measuring walkability, Walk Score, was created for home buyers), walkable neighborhoods have gone up in value and have become less accessible. Areas with higher walkability scores attract higher income people and have higher rates of gentrification (Gibbons and Appleyard, 2025). I can see evidence of displacement when comparing the original HOLC redlined neighborhoods in Figure 2 above with the 2020 neighborhoods by percent of people of color in Appendix C. None of the originally redlined neighborhoods falls in the most POC category with a percentage higher than 41% people of color. If communities most harmed by redlining are no longer the majority residents of these neighborhoods, then higher walkability scores represent a benefit received by newer, whiter, and higher income residents rather than the communities that endured decades of disinvestment. The relationship of walkability and gentrification is remarkably similar to the relationship between redlining and white suburban sprawl. Both cases entail black people being confined to certain spaces to make way for white people's changing desires. Although city planning policies have dropped overtly racist motivations, this study shows a need for policies to include protections for people of color so that historical patterns of oppression do not continue to be repeated.

The findings from Model 3 of this research support the notion that people of color are not the ones benefiting most from high walkability in redlined neighborhoods. The model showed that after controlling for HOLC grades, the percent of POC predicts lower walkability. In other words, areas with the same grade have lower walkability if there are more people of color. This finding was not significant when controlling for vulnerability. Model 3's results are consistent with the pattern in which black and minority communities face walkability deficits that persist independently of both historical policy designation and current economic conditions, suggesting

that race remains an independently meaningful predictor of inequality in Portland's built environment.

The results of this study are entirely dependent on the unique history and structure of Portland. While the results show that previously redlined neighborhoods tend to have the highest walkability scores, we cannot assume that this would be reflected in other cities. The City of Portland has worked consistently to prioritize walkability for decades, and in many more recent years, has continued efforts to atone for racist policy that has disenfranchised black people and allowed for their continued oppression. PedPDX, a 2019 citywide plan to "make Portland a great walking city for everyone", is the latest edition of their groundbreaking 1998 masterplan that started an era of city walkability plans. PedPDX lists equity as its first priority, specifically mentioning how inadequate pedestrian infrastructure disproportionately impacts people of color. Equity was not mentioned in the 1998 version, showing that Portland has made recent specific efforts to improve walkability in areas that were historically discriminated against. PedPDX includes many actions such as the 'Walking while Black Focus group' which is an organized conversation meant to involve black Portlanders so that their issues are specifically addressed. Looking forward, the City of Portland is helping by investing in Albina Vision Trust (AVT), a community-based organization that plans to redevelop the Albina neighborhood that was previously destroyed by Portland's highway and stadium development. AVT's mission is to "ensure that our most marginalized communities are reflected in the built environment" and start the process of building generational wealth through home ownership (2019). Portland has hosted community events allowing for feedback, updates, and review of AVT's work from local stakeholders. The collection of these efforts by the city to repair the harm they caused by centering marginalized people in the retribution process could explain the relationship between

walkability and redlining that this study found. It is probable that walkability rates in marginalized communities are rising and will continue to improve with these efforts.

An alternative explanation for the results found in my analysis is the limitations of the walkability index itself. As previously discussed, measurements and definitions of walkability are broad; the EPA defines walkability as simply as a pace that is "easy to walk around (Thomas and Zeller, 2021, 1)". This matters because the choice of walkability index could define the conclusions that are reached. Like most available measurements of walkability, this study uses an index which measures proximity to amenities, but not their quality. It is possible that while redlined neighborhoods have a higher density, the amenities are worse than those in less dense areas. Quality and perception of neighborhoods could play a large role in walkability, because even if certain amenities are available within walking distance, if people don't feel safe to walk or would prefer to travel to a higher quality school or grocery store, the area should not actually be considered walkable. In this study, formerly redlined areas score higher on walkability by metric, yet people of color within these areas still predict lower walkability scores. That gap suggests that the index does not capture how walkability is experienced differently across racial groups. A true measure of walkability would incorporate community-level perceptions of safety and quality along with measurements of the built environment.

Another takeaway from this research is that walkability should never be considered in a vacuum. While this paper found a connection between redlining and walkability, there are many other factors that influence the health and social outcomes we associate with walkability. While people of color have been displaced due to gentrification, they are still clustered in the dense inner city rather than the outer suburbs and exurbs which tells us that walkability alone is not enough to predict the health of a community, as previously established, POC still have worsened

health outcomes (Zakeri er al., 2024). People in car-dependent suburbs might walk fewer steps than people in walkable city streets, but if the car dependents are able to afford a gym membership, and top-quality healthcare that is inaccessible to others, they will still have a healthier cardiovascular system. Walkability is one piece to a large complex social puzzle. As cities move to create more walkable spaces to improve the quality of life within their streets, they must continue efforts to improve health care, education, job opportunities, and more for their disadvantaged communities. At the end of the day, the positive impact of walkability will not outweigh the harm caused by systemic oppression on its own.

Having a statistically significant relationship between redlining and walkability tells me that overt systemic racism is actively shaping the way walkability exists in our cities today. In 1968, when housing discrimination was made illegal with the FHA, it included the affirmatively furthering fair housing statute (AFFH) (The Fair Housing Act, 2025). This statute mandates that the Department of Housing and Development do more than not discriminate; they must actively work against previous systems of discrimination. The 2015 Rule, coming directly out of the AFFH, requires cities to "take meaningful actions that… address significant disparities… replacing segregated living patterns with truly integrated and balanced living patterns." Given this law, the city of Portland's actions to address racial inequality in walkability planning are not going above and beyond, it is the city fulfilling its legal duty to repair past harm.

The findings in this study offer a direct response to the paradox identified in the literature: if minorities are more likely to live in dense, urban areas, why do they continue to experience worse health outcomes than their white counterparts? First, within similarly graded neighborhoods, higher concentrations of people of color predict lower walkability scores which suggests that walkability metrics fail to capture the unequal ways walkability is experienced by

race. Second, the communities originally harmed by redlining no longer live in formally redlined neighborhoods which could mean that as walkability becomes more desirable, it increases the value of those neighborhoods which pushes out minorities through the process of gentrification. Together, these findings indicate that walkability, as currently defined and measured, does not distribute its benefits equitably. The health disparities documented in the literature are not a contradiction to walkability, but evidence that marginalized communities continue to be systemically excluded from accessing those desirable elements of city living.

Overall, significant questions remain about the complex relationship between historical racist policy and contemporary walkability outcomes. This study provides a spatially grounded, historical background in Portland, Oregon that can help explain modern patterns we see today. The goal of future research should be to create a built environment that provides for the health and well-being of all people, equitably.

Conclusion

Nearly a century after the Home Owner's Loan corporation drew maps that would inform decades of federal decision making, the legacy of their decisions remains present in the makeup of cities today. This study set out to test whether HOLC redlining is visible in the walkability of Portland neighborhoods. While the findings did not confirm the hypothesis in its expected direction, the analysis found a consistent and statistically significant negative relationship between HOLC grade and higher walkability. Rather than suggesting that redlining benefited affected neighborhoods, the findings point to a more nuanced view of walkability. The dense, mixed-use characteristics of inner-city neighborhoods, the same features that made them targets of discriminatory appraisal, are the same characteristics idolized for walkability today. Redlining

caused harm through displacement, wealth stripping, and financial insecurity, not through physical destruction of the urban form which we understand today as walkable.

Formerly redlined neighborhoods being more walkable does not mean that communities originally harmed by redlining are the ones receiving the benefits of walkability. Direct city planning and indirect gentrification has displaced many people of color from their historical neighborhoods, so that they cannot benefit from the positives of modern urban density. Furthermore, this study found evidence that within similarly graded neighborhoods, people of color are associated with lower walkability scores which means that walkability is not a neutral metric and people of color do not have equal access to it.

The 2010 story of creating a bike lane through Portland's Albina neighborhood illustrates the tension that has been growing between vulnerable communities and the city government for decades. PBOT made a mistake by announcing its development plan without an honest effort to involve the community in a decision that would shape the built environment in the heart of Portland's strongest center of black culture. That initial failure was found to be responsible for most of the opposition the city faced (Miller and Lubitow, 2015). In response to the backlash, Portland increased outreach to black cultural centers, involved community members on a decision-making board, and heard feedback at public forums. These changes gave the black community autonomy and helped produce a culturally sensitive compromise that included bike safety, pedestrian protection, and a historical acknowledgement of the neighborhood's importance to black Portlanders. The lesson to be learned from this story is the value of recognizing a space's history when planning for the future.

Further research should investigate how redlining, and other racist historical policies, are related to walkability so it can be determined where cities need to incorporate equity and

reparations to their walkability policy. This study should be replicated in other American cities to assess whether the relationship between redlining and walkability found is the same nationally or specific to Portland. A multi-city comparative study would allow researchers to determine the value of Portland's interventions and support their implementation in other cities. Another area for future work is longitudinal research tracking walkability scores over time. Tracking changes over time would help identify if walkability in formerly redlined neighborhoods is rising due to positive Portland city policies that could be replicated in other areas. Conversely, if walkability in those neighborhoods has been consistently high since before equity efforts, it would suggest that the structural characteristics of these formerly redlined areas are the primary driver of the observed relationship. Additionally, it would be revealing to assess walkability through quality and community perception. A future study should investigate if the pattern of redlined neighborhoods having higher walkability scores stays consistent when using a walkability metric that incorporates quality of amenities and other subjective measures.

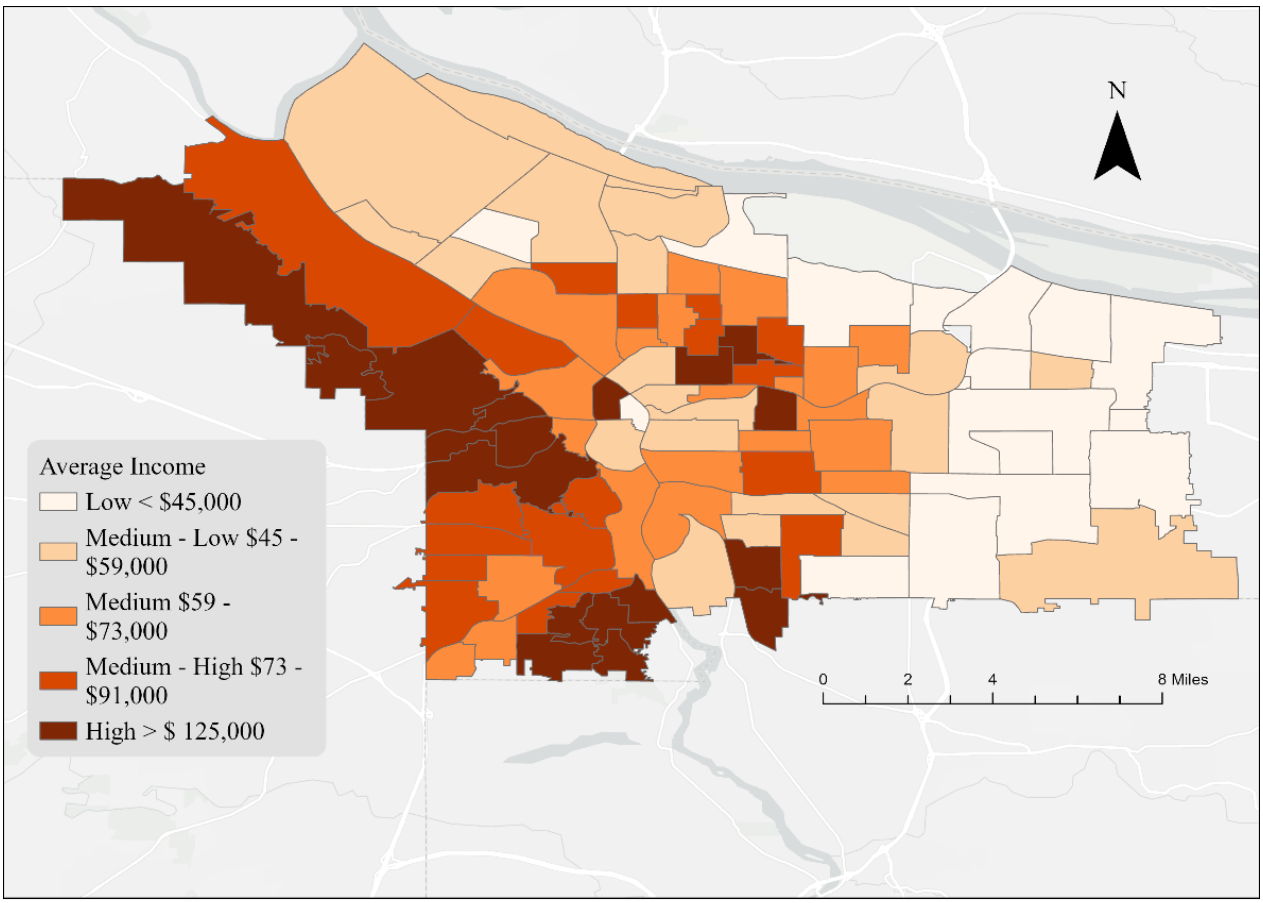
Portland is paving the way for a more equitable style of walkability. The Affirmatively Further Fair Housing statute and its 2015 rule already provide a legal foundation for cities to take similar actions to ensure equity in city planning, whether comparable efforts are being made in other American cities or whether Portland's interventions are sufficient in closing the gap in walkability documented here, are questions future research must address. The benefits of walkability can only be realized when they are ensured for all people. A walkable city is not an inherently equitable one, but with intentional planning that centers the communities most harmed by historical policy, it can be.

Appendices

Appendix A: Acronyms Table

Acronym	Full Name
HAP	Housing Authority of Portland
PBOT	Portland Bureau of Transportation
HOLC	Home Owner's Loan Corporation
FHA	Federal Housing Act
AVT	Albina Vision Trust
EPA	Environmental Protection Agency
AFFH	Affirmatively Furthering Fair Housing Statute

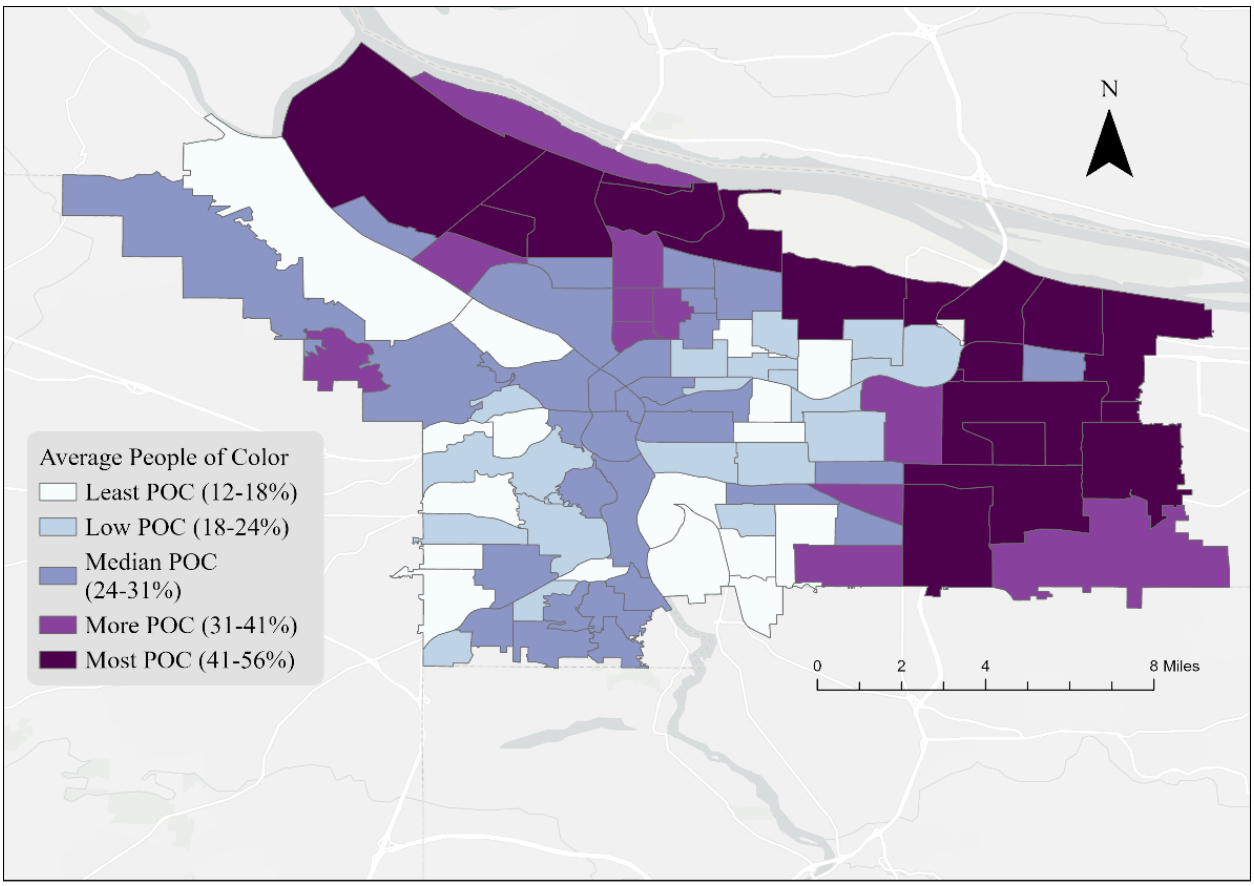
Appendix B: Income Map



Portland, Oregon Average Income

Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

Appendix C: People of Color Spread Map



Portland, Oregon Average POC

Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

Appendix D: GitHub repository containing R Code for data analysis

https://github.com/b-woodbury05/portland-walkability-redlining

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AI Statement

I used AI to find data sources for walkability in Portland. Claude suggested using an overlay from the city of Portland that defined neighborhood opportunities which led me to find my complete neighborhoods data set.

I also used AI during the ArcGIS portion of the study. I created my own workflow, but I am new to using ArcGIS, so I asked Claude for help finding the explicit functions I needed. For example, I asked “How do I join a column from a table to a layer in ArcGIS?”